

Profile

Joined CSK (now SCSK) and was responsible for process management at the steelmaking plant of a major steel manufacturer. After leaving CSK, worked mainly as an independent professional with manufacturers, system integrators, software houses, and users. Responsible for system planning, management, standardization, and design in financial services, including non-bank finance, credit, securities, and electronic money. Also responsible for outsourcing business planning and management, comprehensive system planning and management for a major retailer, including hardware, networks, and systems, preparation of RFPs, vendor selection and management, internal control development, and quality improvement in system development projects.

Served as a part-time lecturer at Kyushu Electric Technical School until 2002. Since 1992, conducted research into system development processes, contract theory, development technologies, organizational theory, and IT evaluation systems in order to realize performance-based compensation.

Performance-Based System Development IT Investment Assessment Business Plan

Transition from Man-Month Billing and Labor-Intensive
Business to Knowledge-Intensive Business

Transformation toward Organizations that Create Innovation

March 24, 2022
Hirano System Design Office

1. Overview of the Performance-Based System Development Business

1-1. Business Description

- Rather than billing based on the traditional man-month model, use the improvement in customer profit generated by system investment as the basis for billing.

Revenue from IT investment assessment consulting fees and fees for providing IT investment assessment tools.

1-2. Business Background

- Japan's international competitiveness has declined relatively, and profitability in the IT industry is lower than in Europe, the United States, and India.

There is a strong possibility that an era will come in which maintaining large-scale systems becomes difficult.

As society rapidly transitions into a digital society, organizations are increasingly likely to need to consider the value of IT utilization.

If the promotion of DX (Digital Transformation) accelerates, IT investment assessment may also be required, similar to ISO 30414, the international standard for human capital disclosure, and the requirements imposed on publicly listed companies in the United States.

The effects of an industry structure centered on contracted development, multi-tier subcontracting, and the delayed transition away from man-month-based business are significant. Measures addressing the declining birthrate and aging population are also delayed. Subcontracted engineers continue to serve as a buffer for economic fluctuations and are becoming exhausted because of deteriorating working environments, employment conditions, and treatment. These people currently support Japan's IT infrastructure.

1-3. Need for the Business

(1) Customer Perspective

Accelerate DX while assessing the balance of IT investment, and increase organizational added value.

(2) Development Perspective

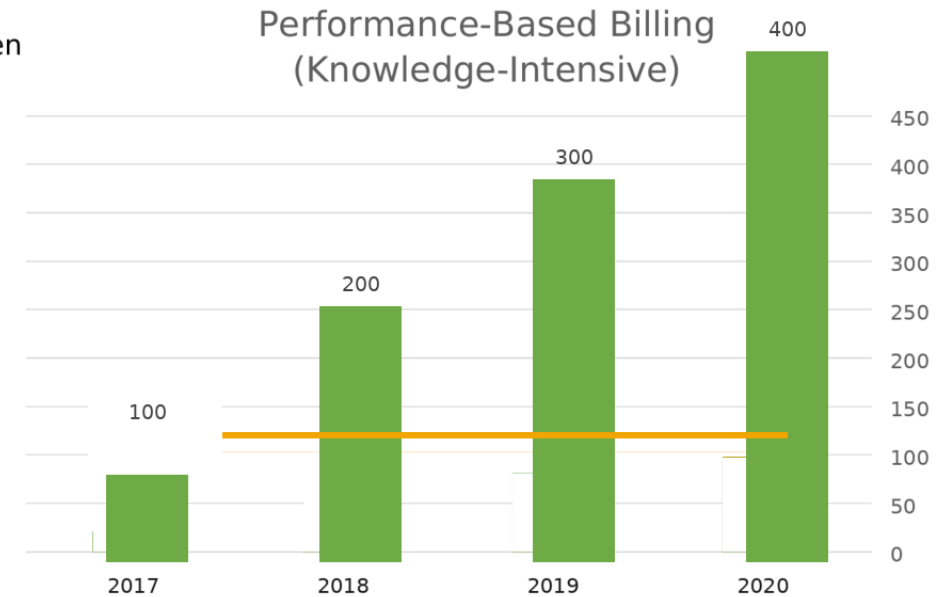
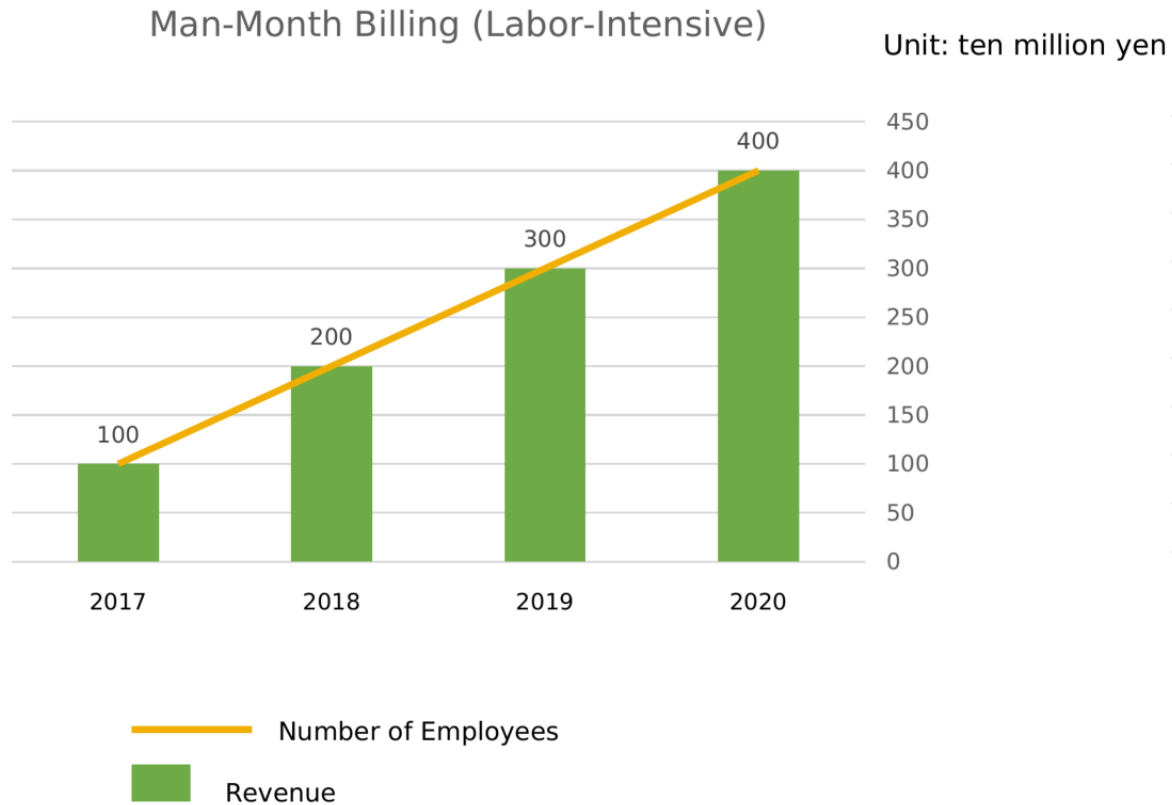
Transform from a labor-intensive model (Man-Month Billing) to a knowledge-intensive model, establish mechanisms for organizational and individual learning, and transform the organization so that it can create a new revenue structure and innovation.

(3) Industry-wide Perspective

Promote the development of industries in Japan and overseas by increasing added value through DX, not only within individual companies but also through collaboration among companies.

2. Objective of the Revenue Structure in Performance-Based System Development

- Revenue is proportional to the number of employees, and a decline in orders can lead to a cash-flow shortfall.
- Customer profit is a factor, so revenue is not proportional to the number of employees.



4. Concept and Verification Policy for Performance-Based System Development Contracts

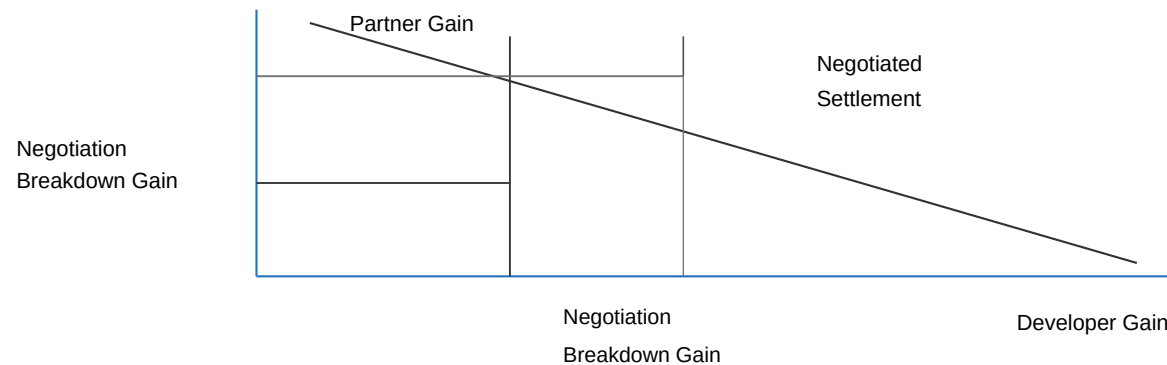
Applying Game Theory and Contract Theory (Overview of Benefit-Sharing Plan)

4-1. Contract Concept

- Prior discussions and a binding agreement are necessary to share the maximized gains.
- A fair settlement point for a partnership with a negotiation breakdown point satisfies the following four conditions:
 - Efficiency
 - Symmetry
 - Invariance
 - Independence of irrelevant alternatives
- Benefit-Sharing Formula

Developer (partner) gain = negotiation breakdown gain

(reference point [ii] α [negotiation condition = contribution]) < total gain



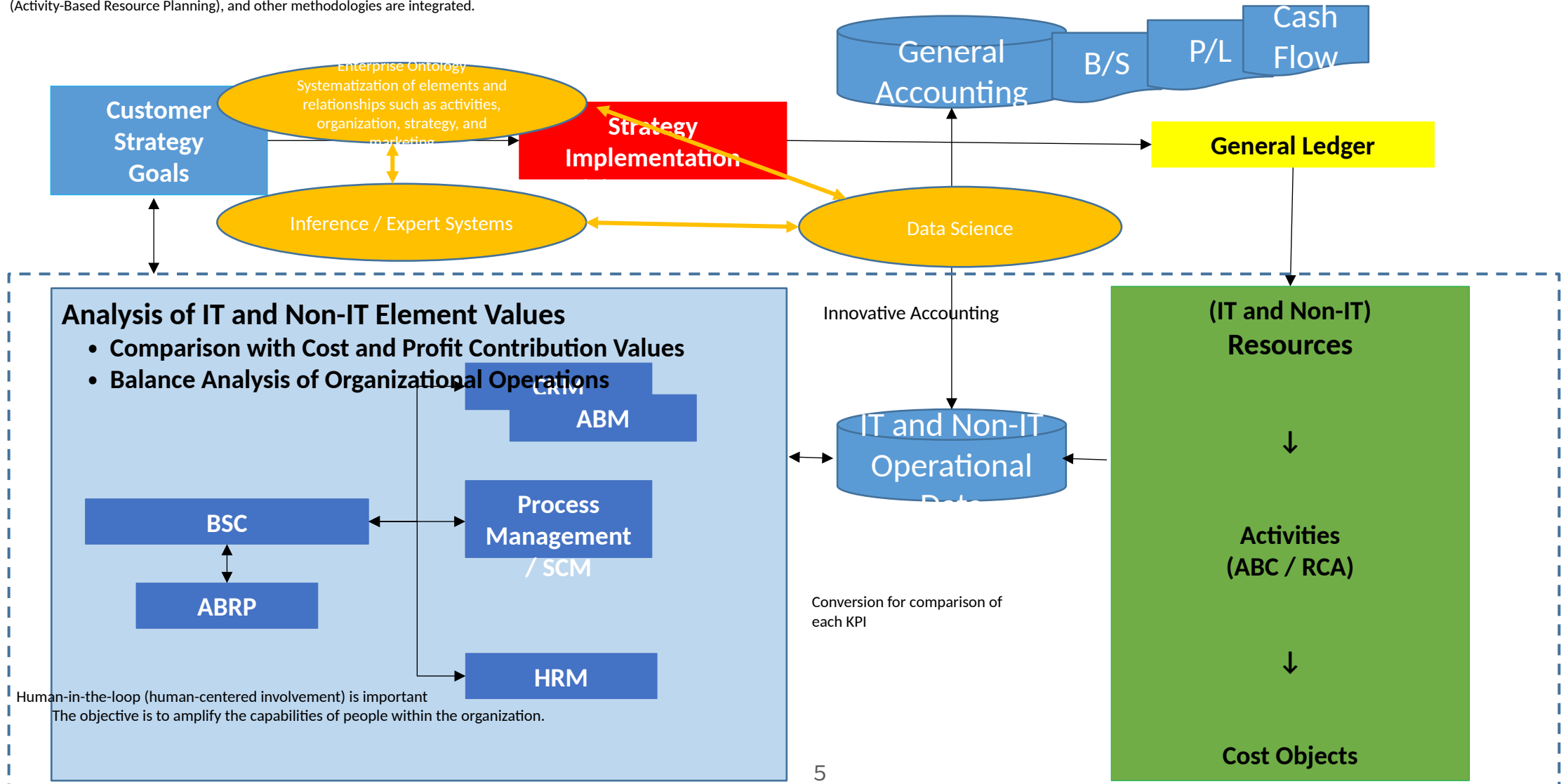
4-2. Effect Verification Policy Based on the Contract Concept

- Clarify what has been a black box within the company, particularly the roles and target figures of IT and non-IT activities.
- Eliminate information asymmetry.
- Clarify the boundaries of the enterprise.

5. Overview of Performance-Based System Development Evaluation Applying Performance Management and Knowledge-Creating Management (Development of Innovative Accounting)

- Relationship between changes in IT and non-IT general ledgers and activities, SCM, CRM, HRM, and profit

* SCM (Supply Chain Management), CRM (Customer Relationship Management), ABM (Account-Based Marketing), HRM (Human Resources Management), BSC (Balanced Scorecard), RCA (Resource Consumption Accounting), ABC (Activity-Based Costing), ABRP (Activity-Based Resource Planning), and other methodologies are integrated.



6. Essential Objectives and Means of Realization for Performance-Based System Development

High Added Value
High Quality
Short Development Time
Low Cost
Safety

Means	Necessity	Details
Human Resources	Software development depends on people who are educated and trained according to the objectives (people produce the results).	Project Management Office Business planners, software engineers, etc.
Methodologies	Software development depends on practical and appropriate processes based on an appropriate organizational structure.	Industrial Engineering, Contract Theory Software Engineering, Ontology Engineering Quality Engineering, Safety Engineering, and other knowledge disciplines
Management	Improve how people can achieve results.	Evaluation Methods Clarification of Measurement Methods
Tools	Improve the quality of human resources, manage work, and systematize work.	Business processes, development, configuration management, testing, deployment, project management, operations management, groupware, etc.
Reuse (Componentization / Patternization)	Low cost, short development time, high productivity, and high quality.	Promote componentization by process and function. Enterprise ontology, design, implementation, and testing
Reengineering	Improve understanding of existing systems and promote maintenance automation	Analysis, restructuring, reverse engineering, and migration

7. Overview of Departmental Plans for the Performance-Based System Development Business

An organization in which people grow, act autonomously with a sense of purpose, engage in skillful dialogue and empathy, and are supported by finance to create innovation

Plan Overview	Basic Policy
Organization Plan	Organize the company into teams by profession rather than by vertically divided industries, and carry out work through task-force-style teams. Based on the principle of complementarity, delegate as much authority as possible to each team (for example, deployment to the production environment, release timing, selection of necessary tools, budget approval, and securing personnel required for projects) to improve efficiency, autonomy, and capability, with a focus on results.
Education and Training Plan	Train people to focus constantly on outcomes rather than merely on results. To achieve this, support self-development plans, encourage regular feedback, provide expenses for training, and allow time to pursue individual goals. Also create mechanisms for learning from failure.
Human Resources System Plan	Separate personnel evaluation from compensation evaluation. Although compensation should be linked to organizational performance evaluation, ranking that combines personnel evaluation and compensation promotes competition rather than cooperation and reduces the integration of organization and individuals and their mutual contribution. Personnel evaluation should be focused solely on improving individual self-development.
Talent Acquisition Plan	Acquire people who are willing to quickly acquire the skills needed for the work, who have their own opinions but are not attached to them, and who, when something cannot be done by themselves, can relinquish leadership authority when necessary to complete the project, even if they are the leader.
Financial Plan	When focusing on outcomes, change from an annual-budget financial cycle to a process flow of financial forecasting, planning, and reporting. Separate funding from the annual budget and consider what should be funded and when. Costs must be carefully managed, but should be linked to business activities while repeating short cycles of planning, forecasting, and monitoring. By limiting resources and time periods, discontinue investments that do not lead to results and redirect resources to exploring new businesses.
Risk Management Plan	Minimize hazards (dangers and obstacles) for customers and the company, work to reduce uncertainty while complying with rules, and give first priority to increasing the possibility of returns by viewing risks as business opportunities.